CS 4400: Artificial Intelligence

Logical Agents

Contents

1. Logical agents
2. The Wumpus environment
3. Logical systems

1/3. Logical agent

- So far in this course: agent finds best action(s) by searching
  - Knowledge about the task: encode by problem formulation
  - Different searching algorithms for partially-observed, stochastic, and/or multiple-agent environments
  - Feasible for small problems
- Next up: agent finds best action(s) by reasoning
  - Knowledge about the task: system for “knowledge representation”
  - Use logical inference algorithms on the “knowledge”
  - Find solution by deriving new “knowledge”

Detective agent

- Trying to solve a murder case
  - Mr. Jones was killed in Old Westbury on Thursday
  - DNA evidence indicates conclusively that either Bob or his twin brother Ron committed the crime
  - Witnesses say Bob was in Hong Kong from Monday through Friday
- Who killed Mr. Jones?
  - *How did we infer this?*

Knowledge

- Explicit knowledge
  - Jones was killed on Thursday
  - Jones was killed in Old Westbury
  - Bob killed Jones or Ron Killed Jones
  - Bob was in Hong Kong Monday through Friday

- Are these enough for an agent to solve the murder?

Knowledge

- Background knowledge
  - You must be in the same place as someone to kill him/her in person
  - A physical object can only be in one place at a time
  - People are physical objects
  - Bob, Ron and Jones are people
  - Thursday is between Monday and Friday
- Humans have very large background knowledge on many things

Logic

- Logic gives us a way to use what we already know to draw conclusions
- Through logic we can also
  - interpret ambiguous perceptions
  - provide compact ways to store knowledge through generalizations (e.g., all men are mortal)
  - transform transient perceptions into long-term memory (knowledge) about the environment
  - infer hidden properties about the environment

Knowledge & Reasoning

To address such logic we will introduce

A knowledge base (KB): a list of facts that are known to the agent.

Rules to infer new facts from old facts using rules of inference.

Logic provides the natural language for this.

Knowledge-based agent

- Knowledge base (KB): what the agent knows about the world so far
- KB is a set of *sentences*
  - Sentences are similar to natural language sentences (e.g., English sentences)
  - Knowledge-representation language (e.g., propositional logic, predicate logic)
- Agent can perform two operations with KB:
  - TELL : put sentences in the KB
  - ASK : query if a sentence is true given the KB

Knowledge-based agent

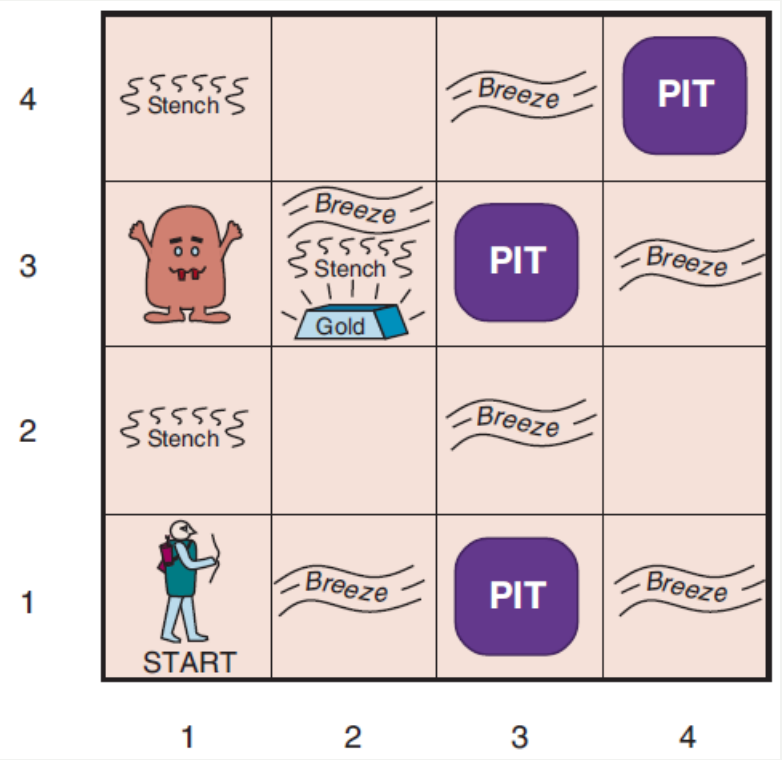
function KB-AGENT(*percept*) **returns** an *action*
  persistent: KB, a knowledge base
    t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(*percept*, t))
  action $\leftarrow$ ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(*action*, t))
  $t \leftarrow t + 1$
  return *action*

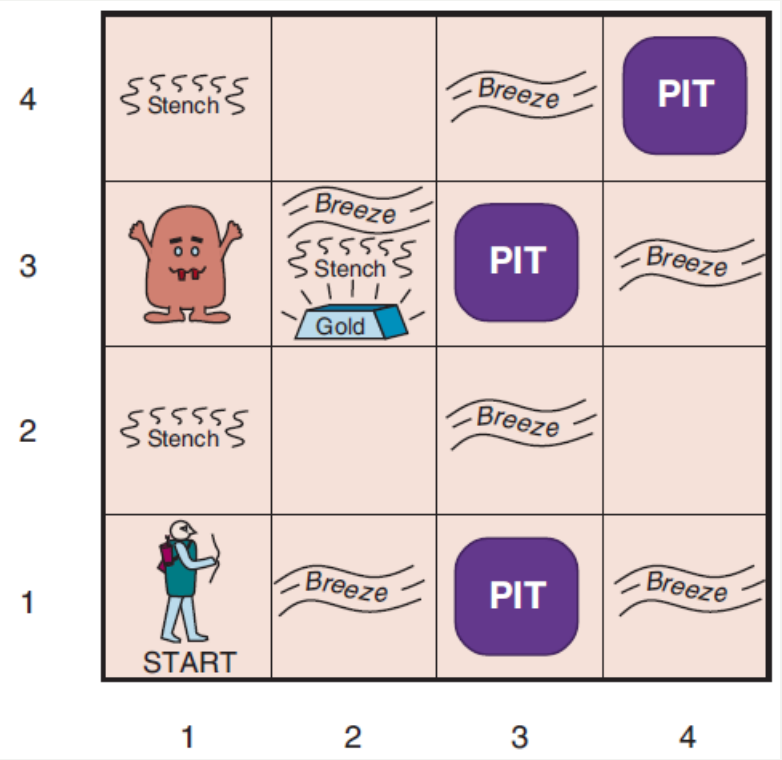
- The KB-agent function is a high-level (=knowledge level) description.
- Implementation level descriptions differ by algorithms.

2/3. Wumpus environment (PEAS)

- Performance measure
  - +1000: get gold; -1000: get killed
  - -1: each action; -10: use arrow
  - Ends when the agent dies or climbs out
- Environment
  - 4 x 4 squares, starting from square [1,1], facing to the right
  - The Wumpus, pits, and gold at unknown locations
  - Instant death if you enter a square with the live Wumpus or a pit
- Actuators
  - Turn 90˚ (CW), Turn -90˚ (CCW), Go
  - Shoot: one arrow to shoot, and shooting the Wumpus kills it



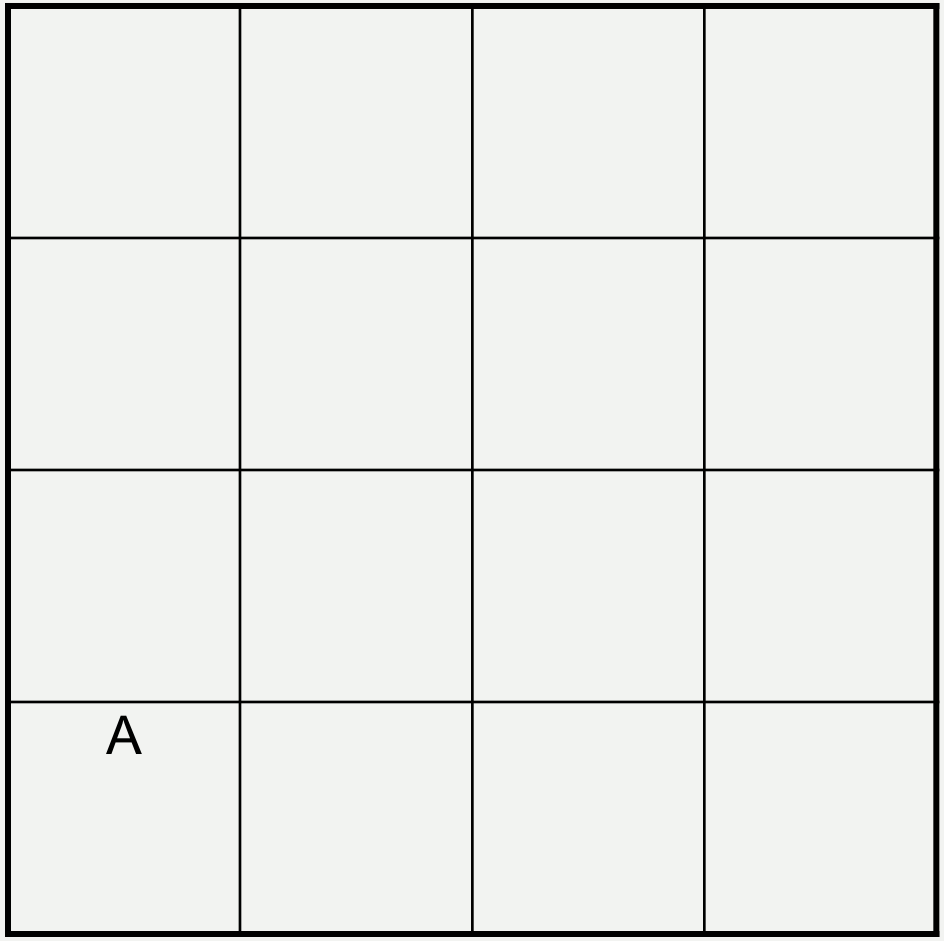
- Actuators (cont'd)
  - Grab: can grab gold if agent is in the same square
  - Can climb out of the cave at square [1,1]
- Sensors
  - No vision – it's too dark to see.
  - Agent perceives a stench in the squares adjacent to the Wumpus
  - Agent perceives a breeze in the squares adjacent to a pit
  - Agent perceives a glitter in the squares with the gold
  - Agent will perceive a bump when it walks into a wall
  - When the Wumpus is killed, it emits a woeful scream that can be heard anywhere in the cave



Wumpus world characterization

Fully Observable No – only local perception
Deterministic Yes – outcomes exactly specified
Episodic No – things we do have an impact.
Static Yes – Wumpus and Pits do not move
Discrete Yes
Single-agent? Yes – Wumpus is essentially a natural feature

Exploring Wumpus environment



- Agent starts in [1,1]
- Perceives:
  [none,none,none,none,none]
(stench,breeze,glitter,bump,scream)

A S OK			
OK	OK		

- Agent starts in [1,1]
- Perceives: [none,none,none,none,none]
- Infers [1,1], [1,2], [2,1] OK
- Moves to [1,2]
- Perceives: [stench,none,none,none,none]

W?			
A S OK	W?		
OK	OK		

- Agent starts in (1,1)
- Perceives: [none,none,none,none,none]
- Infers [1,1], [1,2],[2,1] OK
- Moves to [1,2]
- Perceives: [stench,none,none,none,none]
- Infers: Wumpus in [1,3] or [2,2]

W?			
S OK	W?		
A OK	OK		

- Agent starts in (1,1)
- Perceives: [none,none,none,none,none]
- Infers [1,1], [1,2],[2,1] OK
- Moves to [1,2]
- Perceives: [stench,none,none,none,none]
- Infers: Wumpus in [1,3] or [2,2]
- Infers: only safe move is back to [1,1]
- Moves to [1,1]

W?			
S OK	W?		
OK	A OK		

- Agent starts in (1,1)
- Perceives: [none,none,none,none,none]
- Infers [1,1], [1,2],[2,1] OK
- Moves to [1,2]
- Perceives: [stench,none,none,none,none]
- Infers: Wumpus in [1,3] or [2,2]
- Infers: only safe move is back to [1,1]
- Moves to [1,1]
- Moves to [2,1]

W?			
S OK	W?		
OK	A B OK		

- Agent starts in (1,1)
- Perceives: [none,none,none,none,none]
- Infers [1,1], [1,2],[2,1] OK
- Moves to [1,2]
- Perceives: [stench,none,none,none,none]
- Infers: wumpus in [1,3] or [2,2]
- Infers: only safe move is back to [1,1]
- Moves to [1,1]
- Moves to [2,1]
- Perceives: [none,breeze,none,none,none]

W?			
S OK	W? P?		
OK	A B OK	P?	

- Moves to [2,1]
- Perceives: [none,breeze,none,none,none]
- Infers pit in either [3,1] or [2,2]

W?			
S OK	W?		
OK	A B OK	P	

- Moves to [2,1]
- Perceives: [none,breeze,none,none,none]
- Infers pit in either [3,1] or [2,2]
- BUT! Can't have a pit in [2,2] (no breeze in [1,2])
- Pit MUST be in [3,1]

W			
S OK	OK		
OK	A B OK	P	

- Moves to [2,1]
- Perceives: [none,breeze,none,none,none]
- Infers pit in either [3,1] or [2,2]
- BUT! Can't have Pit in [2,2] (no breeze in [1,2])
- Pit MUST be in [3,1]
- Similarly, wumpus cannot be in [2,2] (no smell in [2,1])
- Wumpus must be in [1,3]

W	OK		
S OK	A OK	OK	
OK	B OK	P	

- Moves to [2,1]
- Perceives: [none,breeze,none,none,none]
- Infers pit in either [3,1] or [2,2]
- BUT! Can't have Pit in [2,2] (no breeze in [1,2])
- Pit MUST be in [3,1]
- Similarly, wumpus cannot be in [2,2] (no smell in [2,1])
- Wumpus must be in [1,3]
- Move to [2,2]
- Perceive: [none,none,none,none,none]
- Infer: [2,3]; [3,2] OK

W	OK		
S OK	A OK	OK	
OK	B OK	P	

- Move to [2,2]
- Perceive: [none,none,none,none,none]
- Infer: [2,3]; [3,2] OK
- Move to [2,3]
- Perceive: [stench,breeze,glitter,none,none]
- Grab gold
- Move back to [1,1]
- Climb out
- Success!

Inference in wumpus environment

- How did we solve the problem?
  - Rules of the environment in KB in the beginning
  - Added new percepts in KB (=TELL)
  - Checked with KB if a square is safe to move in or if it contains gold, etc (=ASK)
  - Assumed that conclusions drawn from KB are guaranteed to be correct if the KB is correct
    - Fundamental assumption for problem solving by logical reasoning
- Let's get into implementing these steps using logical system

3/3. Logical systems

- Components of logical system
  - Representation of knowledge
    - Syntax
    - Semantics
  - Inference
    - World/model
    - Entailment

Knowledge representation

- Syntax and semantics
  - Sentences are expressed according to *syntax* of the representation language
  - Semantics defines the truth of each sentence w.r.t each possible world
- In the language of arithmetic
  - “x + 2 ≥ y” is a sentence; “x2 + y ≥” is not a sentence
  - “x + 2 ≥ y” is true if the number x + 2 is no less than the number y
  - “x + 2 ≥ y” is true in a world where x=7; y =1
  - “x + 2 ≥ y” is false in a world where x=0; y =6

World/model

- A model (= world) : an assignment of values to variables
- Example:
  - x=7; y =1 is a model (= world)
  - x=-1; y=3 is another model (= world)
  - There can be infinitely many possible models

- Q: Is the sentence “x + y = 4” true or not?
- A: It depends on the values of x and y (=model)
- Definitions:
  - A sentence *α* is true in model *m*

    = the model *m* satisfies *α*

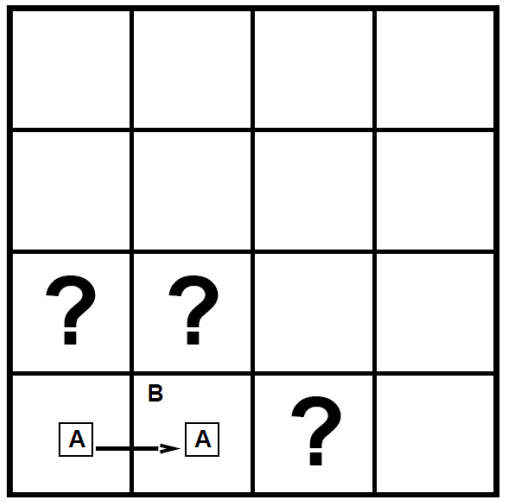
    = *m* is a model of *α*
    - Example: (x=1;y=3) is a model of “x + y = 4” while (x=0;y=2) is not
  - *M*(*α*): all possible models of *α*
    - Example: *M(“x+y=4”)* = { (x=1;y=3),(x=2;y=2),(x=3;y=1) }

Entailment

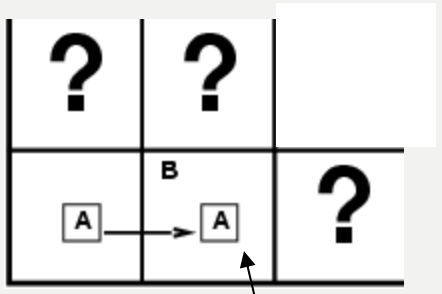
- Entailment
  - KB ⊨ α : the KB entails the sentence α
  - Meaning: sentence α follows logically from the KB
    - Example: “x=0” entails “xy=0”.
- More precisely
  - KB ⊨ α iff every model m of KB is also a model of α
    - Example: for any assignment of x and y such that “x=0” is true, “xy=0” is also true
  - KB ⊨ α iff $M(KB) \subseteq M(\alpha)$
    - Example: M(KB) = {(x=0;y=0), (x=0;y=1),…}
      $\subseteq$ M(α) ={(x=0;y=0), (x=0;y=1), (x=1;y=0), …}

Entailment in wumpus environment

- Consider the situation:
  - Nothing perceived at [1,1],
  - Move right
  - Perceive breeze at [2,1]
- Is it safe to move to [1,2]?
- To answer this, use entailment
  - Consider possible models for the three squares
    - 3 pits, each with 2 assignments (pit or no pit) : 8 possible models



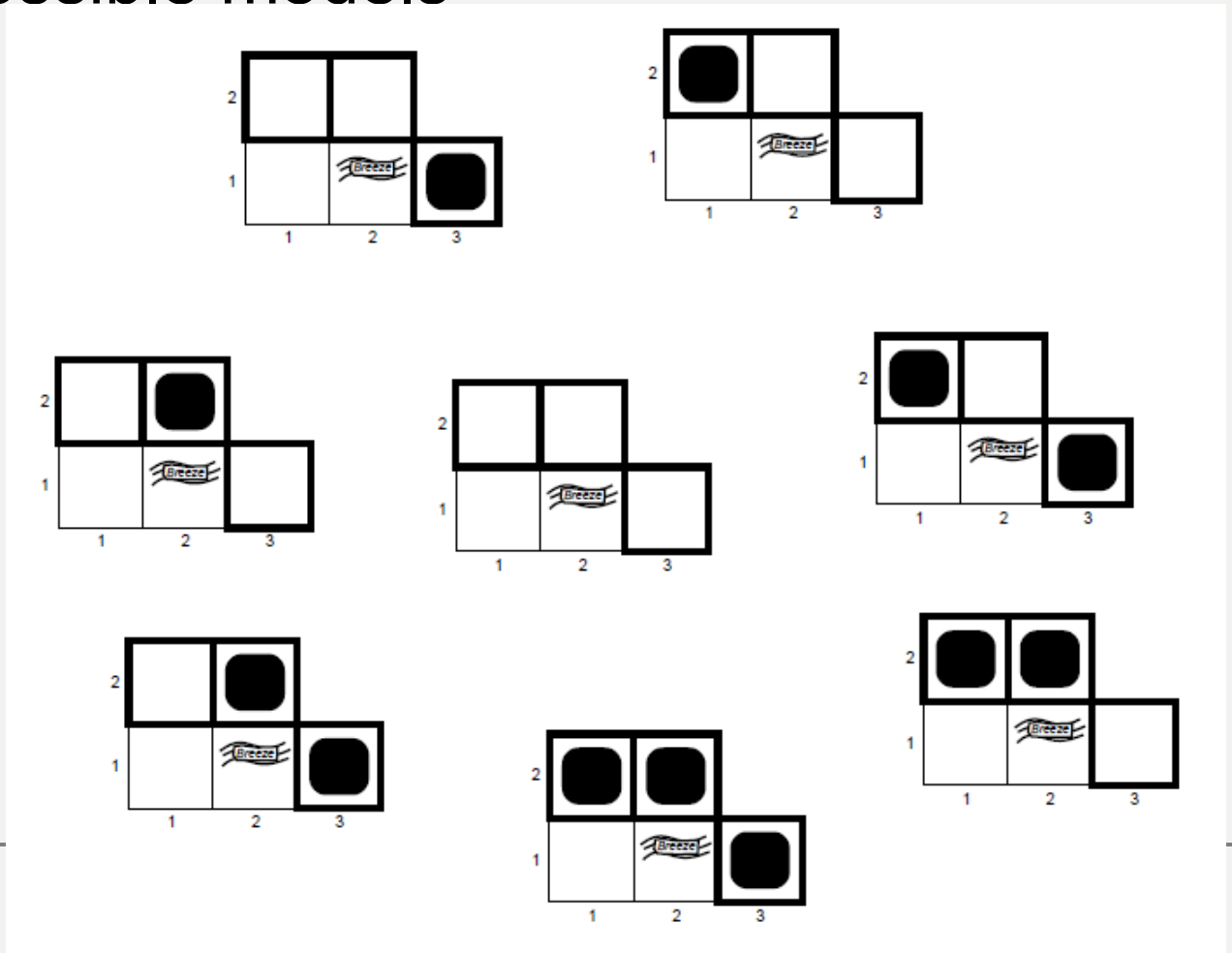
Entailment in wumpus environment

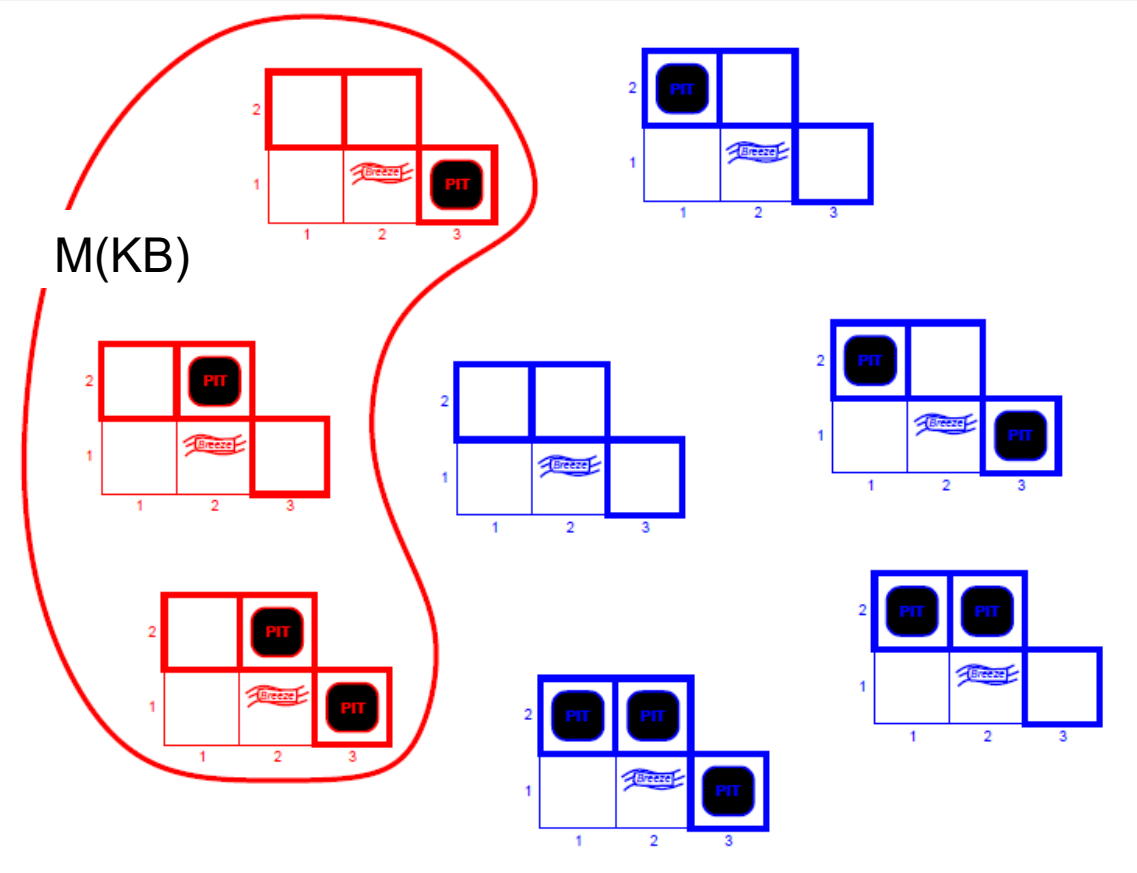


Consider possible models for *KB* assuming only pits and a reduced Wumpus world

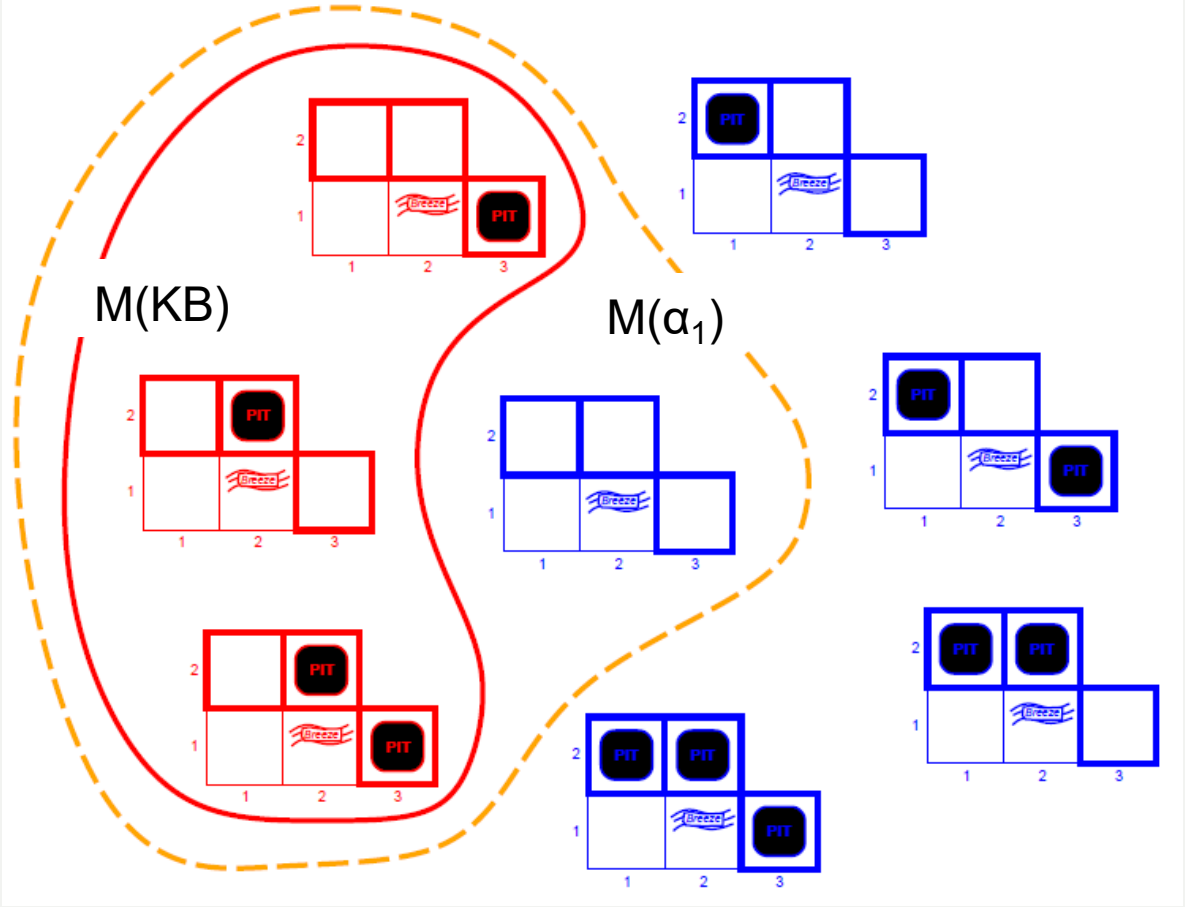
Situation after detecting
nothing in [1,1], moving right, breeze in [2,1]

- 8 possible models

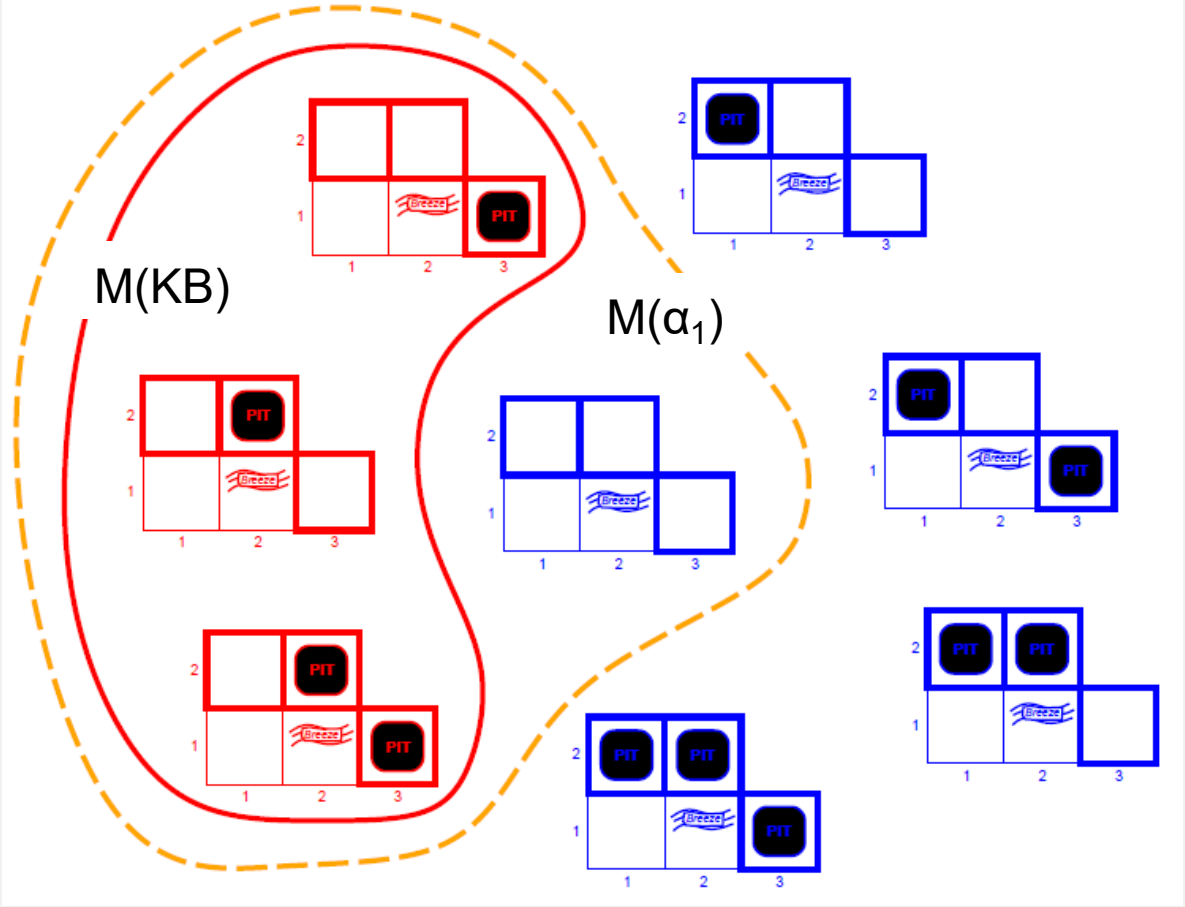




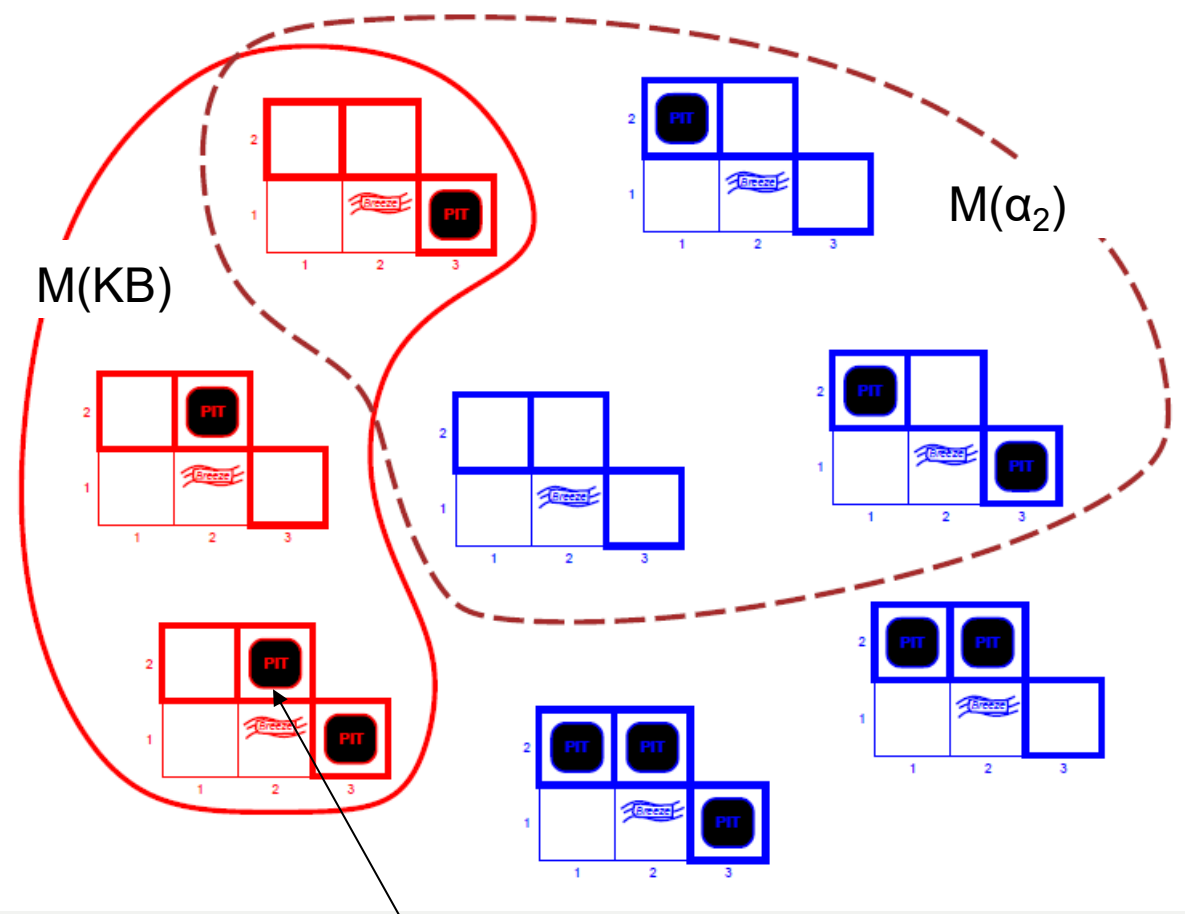
- KB = wumpus-world rules + percepts
  - “Agent feels a breeze in a square adjacent to a pit” + “Agent felt a breeze at square [2,1]”
  - Which model makes the KB true? (=M(KB))



- KB = wumpus-world rules + percepts
- Query sentence α_1 : “[1,2] is safe (has no pit)”
- Which model makes the sentence α_1 true? (=M(α_1))



- Does KB ⊨ "α_1: [1,2] is safe" ?
  - Is every model of KB also a model of α_1?
  - $M(KB) \subseteq M(\alpha_1)$?



- KB = wumpus-world rules + percepts
- Query sentence α_2 : "[2,2] is safe (has no pit)"
- KB ⊨ α_2 ? $KB \not\models \alpha_2$

Inference

- Determining whether KB entails α is the inference procedure
- How did we made inference for Wumpus environment?
  - By enumerating all possible models and comparing M(KB) and M(α)
- There are better algorithms for inference
  - If an inference algorithm i can derive α from KB
  - $KB \vdash_i \alpha$
  - “i derives α from KB” or “α is derived from KB by i ”

Properties of an inference algorithm

- An inference algorithm *i* is sound iff
  - Any sentence derived from KB is entailed by KB
  - = Whenever KB $\vdash_i \alpha$, it is also true that KB $\models \alpha$
- An inference algorithm *i* is complete iff
  - We can derive any sentence that follows from the KB
  - = Whenever KB $\models \alpha$, it is also true that KB $\vdash_i \alpha$
- We need a sound and complete inference:
  - It will answer correctly
  - It will answer any question whose answer follows from what is known by the KB

(Next) Propositional logic

- Propositional logic is one of the simplest and intuitive logical systems (Ch 8)
- There are more powerful logical systems such as first-order logic (Ch 9)

- Propositional logic
  - Syntax: propositions are symbols or combinations of symbols
  - Semantics: propositions are either true or false